SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA0606 COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)

Assessment Tool Weightage: 20%

QUESTIONS: 15 Total Marks: 25 Annexure A

CONCEPT MAPPING

Code of Conduct:

I S.ROHINI / 192424413 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.ROHINI

Annexure A

Short Answer Test

1. Design a machine learning system for disaster management to predict events such as floods and earthquakes using historical and real-time data. Explain the choice ofmodel, data preprocessing, and feature selection techniques.
(a) Describe the steps involved in training, validating, and evaluating the model.
(b) Explain how the system can assist in early warning and emergency response. (5 Marks)

Machine Learning System:

Choice of Model

Random Forest is selected because it provides high accuracy, handles large datasets efficiently, and reduces overfitting.

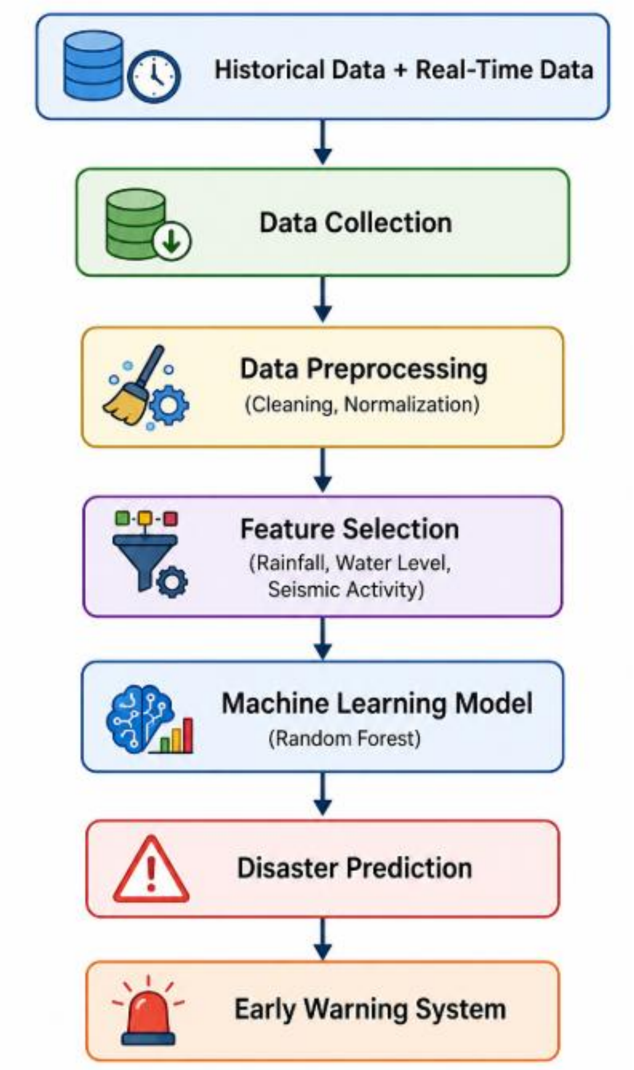
Data Preprocessing

- Remove missing values.
- Remove duplicate records.
- Normalize numerical values.
- Encode categorical data.

Feature Selection

Important features include:

- Rainfall
- Water Level
- Seismic Activity
- Temperature
- Humidity



(a) Training, Validation and Evaluation

Training

- Train the model using historical disaster data.

Validation

- Validate the model using unseen validation data.
- Tune model parameters to improve performance.

Evaluation

- Test the model using test data.
- Evaluate using Accuracy, Precision, Recall and F1-Score.

(b) Early Warning and Emergency Response

- Predict disasters before they occur.
- Send alerts to government authorities.
- Help in evacuation planning.
- Allocate rescue teams efficiently.
- Reduce loss of life and property.

Conclusion

A machine learning-based disaster management system improves prediction accuracy and supports effective emergency response.

2. In the context of the Find-S algorithm, consider the dataset used to determine whether a customer will buy a product based on attributes such as Age, Income, Student, and Credit Rating. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show step-by-step updates after each positive example and ignore negative examples. Finally, provide the final hypothesis. (5 marks)

Dataset:

Example	Age	Income	Student	Credit Rating	Buys Product
1	Young	High	No	Fair	No
2	Young	High	No	Excellent	No
3	Middle	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes

Initial Hypothesis

S0 = Ø

Example 1

Negative example → Ignored

S0 = Ø

Example 2

Negative example → Ignored

S0 = Ø

Example 3

Positive example

S1 = <Middle, High, No, Fair>

Example 4

Positive example

Compare attribute

Attribute	Old	New	Result
Age	Middle	Senior	?
Income	High	Medium	?
Student	No	No	No
Credit Rating	Fair	Fair	Fair

Final Hypothesis

<?, ?, No, Fair>

Conclusion

The maximally specific hypothesis is:

<?, ?, No, Fair>

3. In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a customer will subscribe to a service based on attributes such as Age Group, Income Level, Internet Usage, and Device Type. Determine the version space by computing the specific (S) and general (G) boundaries. Show step-by-step updates after each training example. (5 Marks)

Dataset:

Example	Age Group	Income Level	Internet Usage	Device Type	Subscribe
1	Young	High	Frequent	Mobile	Yes
2	Young	High	Occasional	Mobile	Yes
3	Middle	Low	Frequent	Desktop	No
4	Young	Medium	Frequent	Mobile	Yes
5	Senior	High	Occasional	Desktop	No

Apply the Candidate Elimination Algorithm.

Initial Boundaries

Specific Boundary (S)

<Ø, Ø, Ø, Ø>

General Boundary (G)

<?, ?, ?, ?>

Example 1 (Positive)

S = <Young, High, Frequent, Mobile>

G = <?, ?, ?, ?>

Example 2 (Positive)

S = <Young, High, ?, Mobile>

Example 3 (Negative)

Generalize G

G1 = <Young, ?, ?, ?>

G2 = <?, High, ?, ?>

G3 = <?, ?, Occasional, ?>

G4 = <?, ?, ?, Mobile>

Example 4 (Positive)

S = <Young, ?, ?, Mobile>

Remove inconsistent hypotheses.

Example 5 (Negative)

Final Boundaries

Specific Boundary

<Young, ?, ?, Mobile>

General Boundary

<Young, ?, ?, Mobile>

Version Space

S = G = <Young, ?, ?, Mobile>

Conclusion

The version space indicates that young customers using mobile devices are likely to subscribe regardless of income level and internet usage.

4. Explain the architecture of a typical machine learning pipeline with a neat diagram. Discuss the role of each stage including data collection, preprocessing, feature engineering, model training, evaluation, and deployment. Also, differentiate between training error and generalization error with suitable examples. (5 Marks)

   Machine Learning Pipeline:

Stages

1. Data Collection

- Collect data from various sources.

2. Data Preprocessing

- Remove missing values.
- Remove duplicate records.
- Normalize data.

3. Feature Engineering

- Select important features.
- Create new meaningful features.

4. Model Training

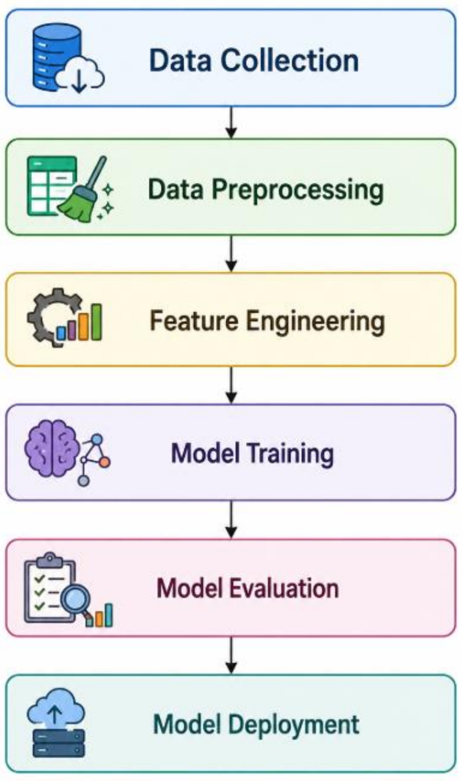
- Train the machine learning algorithm.

5. Model Evaluation

- Evaluate using Accuracy, Precision, Recall and F1-Score.

6. Model Deployment

- Deploy the model for real-time prediction.



Difference Between Training Error and Generalization Error

Training Error	Generalization Error
Error on training data	Error on unseen test data
Measures learning performance	Measures real-world performance
Usually lower	Usually higher

Example

Training Accuracy = **98%**

Testing Accuracy = **93%**

The difference represents the generalization error.

Conclusion

A proper machine learning pipeline improves prediction accuracy and model reliability.

5. Explain the concept of clustering in unsupervised learning. (5 Marks)

(a) Describe the working of the K-Means algorithm with a step-by-step example.

(b) Explain how the value of K is chosen using the Elbow Method.

(c) Compare K-Means with Hierarchical Clustering.

Clustering

Clustering is an **unsupervised learning technique** that groups similar data points into clusters without using labeled data.

(a) K-Means Algorithm

Steps

1. Choose the number of clusters (K).
2. Initialize K centroids.
3. Assign each data point to the nearest centroid.
4. Update the centroid positions.
5. Repeat until centroids remain unchanged.

Example:

● ● ●

● ●

● ●

K = 2

Cluster 1 → Left Group

Cluster 2 → Right Group

(b) Elbow Method

The Elbow Method is used to determine the optimal value of K.

Steps

- Run K-Means for different K values.
- Calculate the Within Cluster Sum of Squares (WCSS).
- Plot K versus WCSS.
- Select the K value where the graph forms an "elbow."

(c) Comparison

K-Means	Hierarchical Clustering
Requires K before training	No need to specify K initially
Fast for large datasets	Slower for large datasets
Centroid-based	Tree (Dendrogram)-based
Produces flat clusters	Produces hierarchical clusters

Conclusion

K-Means is suitable for large datasets with a known number of clusters, while Hierarchical Clustering is useful for exploring relationships among data points without predefining the number of clusters.